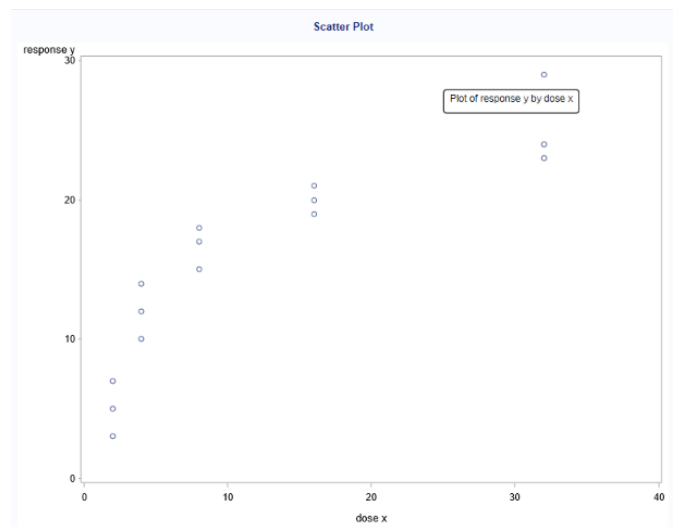


9. a)



Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	590.91613	590.91613	44.28	<.0001
Error	13	173.48387	13.34491		
Corrected Total	14	764.40000			

Root MSE	3.65307	R-Square	0.7730
Dependent Mean	15.80000	Adj R-Sq	0.7556
Coeff Var	23.12089		

Parameter Estimates					
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t
Intercept	1	8.66667	1.42787	6.07	<.0001
dose x	1	0.57527	0.08845	6.65	<.0001

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	2	673.82062	336.91031	44.63	<.0001
Error	12	90.57938	7.54828		
Corrected Total	14	764.40000			

Root MSE	2.74741	R-Square	0.8815
Dependent Mean	15.80000	Adj R-Sq	0.8618
Coeff Var	17.38889		

Parameter Estimates					
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t
Intercept	1	4.48366	1.65720	2.71	0.0191
dose x	1	1.50633	0.28836	5.22	0.0002
xsquared	1	-0.02699	0.00814	-3.31	0.0062

b)

Linear model: $y = \beta_0 + \beta_1 x_1 + e$

Fitted linear model:

$$y = 8.667 + 0.575x_1$$

Adjusted $R^2 = 0.7556$

Quadratic model: $y = \beta_0 + \beta_1 x_1 + \beta_2 x_1^2 + e$

Fitted quadratic model:

$$y = 4.484 + 1.506x_1 - 0.027 x_1^2$$

Adjusted $R^2 = 0.8618$

c) The quadratic equation seems to fit the data better. It has a higher R^2 value and all of its variables are significant in explaining response time at $\alpha = 0.05$

14. a)

$$\text{Model 1: } y = \beta_0 + \beta_1 x_1 + \beta_2 x_1^2 + \beta_3 x_2 + e$$

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	3	63.19591	21.06530	222.41	<.0001
Error	56	5.30393	0.09471		
Corrected Total	59	68.49983			

Root MSE	0.30775	R-Square	0.9226
Dependent Mean	2.77833	Adj R-Sq	0.9184
Coeff Var	11.07696		

Parameter Estimates					
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t
Intercept	1	1.64068	0.09674	16.96	<.0001
additive x1	1	0.00248	0.00415	0.60	0.5518
x1 sq	1	0.00026522	0.00003982	6.66	<.0001
high/low copper x2	1	0.00020500	0.00019885	1.03	0.3085

$$\text{Model 2: } y = \beta_0 + \beta_1 x_1 + \beta_2 x_1^2 + \beta_3 x_2 + \beta_4 x_1 x_2 + \beta_5 x_1^2 x_2 + e$$

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	5	64.22989	12.84598	162.46	<.0001
Error	54	4.26995	0.07907		
Corrected Total	59	68.49983			

Root MSE	0.28120	R-Square	0.9377
Dependent Mean	2.77833	Adj R-Sq	0.9319
Coeff Var	10.12115		

Parameter Estimates					
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t
Intercept	1	1.55764	0.09815	15.87	<.0001
additive x1	1	0.00858	0.00415	2.07	0.0435
x1 sq	1	0.00020478	0.00004145	4.94	<.0001
high/low copper x2	1	0.00036419	0.00025167	1.45	0.1536
interaction	1	0.00232	0.00099558	2.33	0.0238
x1 sq h/l copp	1	-2.30981E-7	9.980404E-8	-2.31	0.0245

The models appear to have very similar (and high) adjusted R^2 values, but the second has a slightly higher one. This model also has less “superfluous” variables according to the p-values of the parameter estimates.

b) Model 1 predicted equation: $y = 1.64068 + 0.00248x_1 + 0.000265x_1^2 + 0.000205x_2$

Model 2 predicted equation:

$$y = 1.55764 + 0.00858x_1 + 0.000205x_1^2 + 0.0003641x_2 + 0.00232x_1x_2 + 0.00000000231x_1^2x_2$$

c) β_1 represents the rate of increase in feed efficiency ratio. That is, if the amount of additive increased by 1, the feed efficiency ratio would increase by 0.00858 in the second model.

23.

a) If a model had an R^2 value of 58.2%, it means that the variables present in the model explain 58.2% of the variance in maximal oxygen uptake. Generally, the higher the R^2 value, the better “fit” or representation it is for the sample data, i.e. a higher percentage of variability explained means there is less random error and any outliers are better accounted for.

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	8	6.21761	0.77720	8.17	<.0001
Error	45	4.28239	0.09516		
Corrected Total	53	10.50000			

Root MSE	0.30849	R-Square	0.5922
Dependent Mean	2.00000	Adj R-Sq	0.5196
Coeff Var	15.42435		

Parameter Estimates					
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t
Intercept	1	-4.27528	14.31481	-0.30	0.7666
weight x1	1	0.05155	0.03797	1.36	0.1814
age x2	1	0.38198	1.22584	0.31	0.7568
walk x3	1	0.05313	0.54684	0.10	0.9230
heartrate x4	1	-0.00501	0.02717	-0.18	0.8546
x1 sq	1	-0.00012811	0.00012507	-1.02	0.3112
x2 sq	1	-0.01073	0.02900	-0.37	0.7132
x3 sq	1	-0.00650	0.01703	-0.38	0.7044
x4 sq	1	-0.00001615	0.00010167	-0.16	0.8745

b) R^2 increased by 1.02 percentage points.

c) None appear to be significant at 0.05 level, all p-values are greater than 0.05

d) All of the p-values for the squared terms are much greater than 0.05, so they would not be significant if taken as a group of four terms.

e) Although the model with second-order terms has a higher R^2 value, all of the higher order terms in the model are insignificant at any realistic/appropriate level of significance. This means that their inclusion is not necessary, so the model with the first-order terms only would probably be recommended.

24.

a)

First-order only model $R^2 = 0.5812$, adjusted = 0.5470

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	4	6.10232	1.52558	17.00	<.0001
Error	49	4.39768	0.08975		
Corrected Total	53	10.50000			

Root MSE	0.29958	R-Square	0.5812
Dependent Mean	2.00000	Adj R-Sq	0.5470
Coeff Var	14.97903		

Parameter Estimates					
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t
Intercept	1	5.58202	1.03103	5.41	<.0001
weight x1	1	0.01292	0.00283	4.56	<.0001
age x2	1	-0.08317	0.03486	-2.39	0.0209
walk x3	1	-0.15789	0.02860	-5.54	<.0001
heartrate x4	1	-0.00910	0.00251	-3.63	0.0007

First-order and second-order model R^2 (from #23) = 0.5922, adjusted = 0.5196

First-order and second-order with interaction terms model $R^2 = 0.6263$, adjusted = 0.4908

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	14	6.58558	0.46997	4.65	<.0001
Error	39	3.91442	0.10038		
Corrected Total	53	10.50000			

Root MSE	0.31762	R-Square	0.6263
Dependent Mean	2.00000	Adj R-Sq	0.4908
Coeff Var	15.88101		

Parameter Estimates					
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t
Intercept	1	7.82668	25.14270	0.31	0.7571
weight x1	1	-0.02046	0.10011	-0.20	0.8391
age x2	1	-0.80677	1.67357	-0.48	0.6325
walk x3	1	0.06057	0.87006	0.08	0.9398
heartrate x4	1	0.08529	0.11291	0.76	0.4546
x1 sq	1	-0.00017005	0.00013496	-1.26	0.2142
x2 sq	1	-0.00300	0.03094	-0.10	0.9233
x3 sq	1	-0.02196	0.02103	-1.04	0.3028
x4 sq	1	-0.00015444	0.00016950	-0.91	0.3678
x1x2	1	0.00287	0.00344	0.84	0.4087
x1x3	1	0.00078973	0.00253	0.31	0.7599
x1x4	1	0.00007051	0.00029496	0.24	0.8123
x2x3	1	0.03433	0.02487	1.38	0.1753
x2x4	1	-0.00095200	0.00259	-0.37	0.7157
x3x4	1	-0.00272	0.00258	-1.05	0.2981

b) None of the cross-product terms provide a significant relationship with y at $\alpha=0.05$, as all of their p-values are greater than 0.05

c) The third model has the highest R^2 value, but it also has the lowest adjusted R^2 value, implying it has “unnecessary variables”. For that reason, the best model may be the first model, as it has the highest adjusted R^2 and the lowest MSE of the three models.

25. Model 4

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	5	63.19920	12.63984	129.34	<.0001
Error	54	5.27737	0.09773		
Corrected Total	59	68.47657			

Root MSE	0.31262	R-Square	0.9229
Dependent Mean	2.77850	Adj R-Sq	0.9158
Coeff Var	11.25127		

Parameter Estimates					
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t
Intercept	1	1.65957	0.12671	13.09	<.0001
additive x1	1	0.00267	0.00596	0.45	0.6564
x1 sq	1	0.00025786	0.00005720	4.51	<.0001
copp x2	1	0.00011994	0.00044799	0.27	0.7904
x1x2	1	-0.00000106	0.00002107	-0.05	0.9601
x1 sq x2	1	3.794843E-8	2.02243E-7	0.19	0.8519

a) Only the additive squared coefficient (beta 2) is significant at $\alpha=0.05$, since it is the only parameter estimate with a p-value less than 0.05

Model 3

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	3	63.17480	21.05827	222.43	<.0001
Error	56	5.30176	0.09467		
Corrected Total	59	68.47657			

Root MSE	0.30769	R-Square	0.9226
Dependent Mean	2.77850	Adj R-Sq	0.9184
Coeff Var	11.07403		

Parameter Estimates					
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t
Intercept	1	1.64133	0.09672	16.97	<.0001
additive x1	1	0.00245	0.00415	0.59	0.5565
x1 sq	1	0.00026545	0.00003981	6.67	<.0001
copp x2	1	0.00020583	0.00019951	1.04	0.3045

b) $H_0: \beta_4 = \beta_5 = 0$ vs. H_1 : at least one inequality

$$F = \frac{\frac{SSR_1 - SSR_2}{3}}{SSE_1/df_1} = \frac{63.1992 - 63.1748}{0.09773} = 0.08322 \text{ test statistic}$$

Critical value: $F_{0.05, 3, 56} = 2.769$

Test statistic is less than critical value, meaning it does not fall within the rejection region. Thus, we would not reject the null hypothesis and conclude that the extra parameters are not significant, meaning Model 4 does not provide a significant gain over model 3

c) The results seem to be consistent as the p-values for both interaction term parameters (β_4, β_5) are *much* greater than 0.05, and are in fact much larger than any useful level of significance, meaning it is very difficult to reject the null hypothesis that the parameters both equal 0. Model 3 also has a slightly higher adjusted R^2 value at 0.9184 compared to model 4's 0.9158, along with a lower mean squared error (0.09467 and 0.09773, respectively), suggesting it is a better predictor of feed efficiency (y). The normality of errors also holds for model 3, so there is no violation of underlying assumptions.

26.

a)

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	58.86420	58.86420	355.18	<.0001
Error	58	9.61236	0.16573		
Corrected Total	59	68.47657			

Root MSE	0.40710	R-Square	0.8595
Dependent Mean	2.77850	Adj R-Sq	0.8572
Coeff Var	14.65179		

Parameter Estimates					
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t
Intercept	1	1.32857	0.09317	14.26	<.0001
additive x1	1	0.02900	0.00154	18.85	<.0001

Model 1 adjusted R^2 : 0.8572

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	2	58.96588	29.48294	176.70	<.0001
Error	57	9.51068	0.16685		
Corrected Total	59	68.47657			

Root MSE	0.40848	R-Square	0.8611
Dependent Mean	2.77850	Adj R-Sq	0.8562
Coeff Var	14.70138		

Parameter Estimates					
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t
Intercept	1	1.28740	0.10734	11.99	<.0001
additive x1	1	0.02900	0.00154	18.78	<.0001
copp x2	1	0.00020583	0.00026367	0.78	0.4382

Model 2 adjusted R^2 : 0.8562

Model 3 adjusted R^2 : 0.9184

Model 4 adjusted R^2 : 0.9158

b) See 25b) which was a very similar procedure leading to the same conclusion. Said conclusion was not to reject the null hypothesis, as the cross-product terms were both greater than the significance level of 0.05

c) I would recommend the third model as it has the highest adjusted R^2 value and the lowest MSE, meaning it provides a relatively strong fit compared to the other models.